

SQL AGGREGATE FUNCTION & OPERATORS

Table students

Student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	FINANCE
5	Eve	22	HR

01 List all distinct department in the student table

```
SELECT DISTINCT department
FROM students;
```

department
IT
HR
finance

02 Get the average age of student per department

```
SELECT department, AVG(age) AS avg-age
FROM students
GROUP BY department
```

department	avg-age
IT	21
HR	22
finance	23

Q3 Show department with more than 1 student

```
SELECT department, COUNT(*) AS student_count  
FROM Students  
GROUP BY departments  
HAVING COUNT (*) > 1;
```

department	student_count
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IT	2
----	---

HR	2
----	---

Q4 Get all students whose age is between 21 and 23

```
SELECT Student_id, name, age, department  
FROM Students GROUP BY department  
WHERE age BETWEEN 21 AND 23;
```

Student_id	name	age	department
2	Bob	22	HR
4	Biana	23	Finance
5	Eva	22	HR
3	Charlie	21	IT

Q5 List all students in the IT ~~and~~ HR department who are older than 21

```
SELECT Student_id, name, age, department  
FROM Students  
WHERE department IN ('IT', 'HR')  
AND age > 21
```


student_id	name	age	department
2	Rob	22	HR
5	Eve	22	HR

Table courses

Course_id	Course_name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

06 Show total credits per department, only for departments with more than 5 total credits

```
SELECT department, SUM(credits) AS total_credits
FROM Course
GROUP BY department
HAVING SUM(credits) > 5;
```

department	total_credits
IT	11

07 List all course that do not have 4 credits

```
SELECT course_id, course_name, department, credits
FROM course
WHERE credit != 4;
```


Course_id	Course_name	department	credits
101	SAL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

08 show the top 3 courses by credits in descending order.

```
SELECT course_id, Course_name, credits
FROM courses
ORDER BY credits Desc
Limit 3;
```

Course_id	Course_name	credits
102	Python	4
103	Data Science	4
101	SAL Basics	3

Take enrollments

enrolment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

09 Get the maximum, minimum and average grade across all enrollments


```

SELECT MAX(grade) AS max_grade
      MIN(grade) AS min_grade
      AVG(grade) AS avg_grade
FROM enrollments;

```

max_grade	min_grade	avg_grade
90	78	85

10 Count how many enrollments exist per course

```

SELECT course_id,
       COUNT(*) AS enrolment_count
FROM enrollments
GROUP BY course_id;

```

Course id	enrolment_count
1	1
2	2
3	3
4	4
5	5

11

Table salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

11 Find total salary and total bonus per department

```
SELECT department,  
       SUM(salary) AS total_salary,  
       SUM(bonus) AS total_bonus  
FROM salaries  
GROUP BY department;
```

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12 Show department where average salary is above 55000

```
SELECT department,  
       AVG(salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61000

13 List employees whose salary plus bonus is greater than 60000

```
SELECT employee_id,  
       name,  
       salary,  
       bonus,  
       (salary + bonus) AS total_compensation  
FROM salaries
```


Where (salary + bonus) > 60000;

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Table projects

project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

14 Show total and average budget per department. Only include departments with average budget above 70000

department	total_budget	avg_budget
1		

```

SELECT department,
       SUM(budget) AS total_budget,
       AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING AVG(budget) > 70000

```

department	total_budget	avg_budget
IT	270000	135000

15 List all projects with budgets between 50000 and 120000
(excluding Marketing)

```
SELECT project_id,  
       project_name,  
       department,  
       budget  
FROM   projects  
WHERE  budget BETWEEN 50000 AND 120000
```

project_id	project_name	department	budget
2	payroll system	Finance	80000